CAPSTONE · THESIS · DATA SCIENCE

Lecture Autosummary

Grounded multi-source lecture summaries
with adaptive comprehension checks

Shamil Arslanov · Harbour Space · 2026
github.com/homomorfism/autosummary

PROBLEM

After a lecture, both artifacts fail the student

Recording

2–3.5 hours per lecture

too long to re-watch

Slides

examples · caveats · intuition ·
“this will be on the exam”

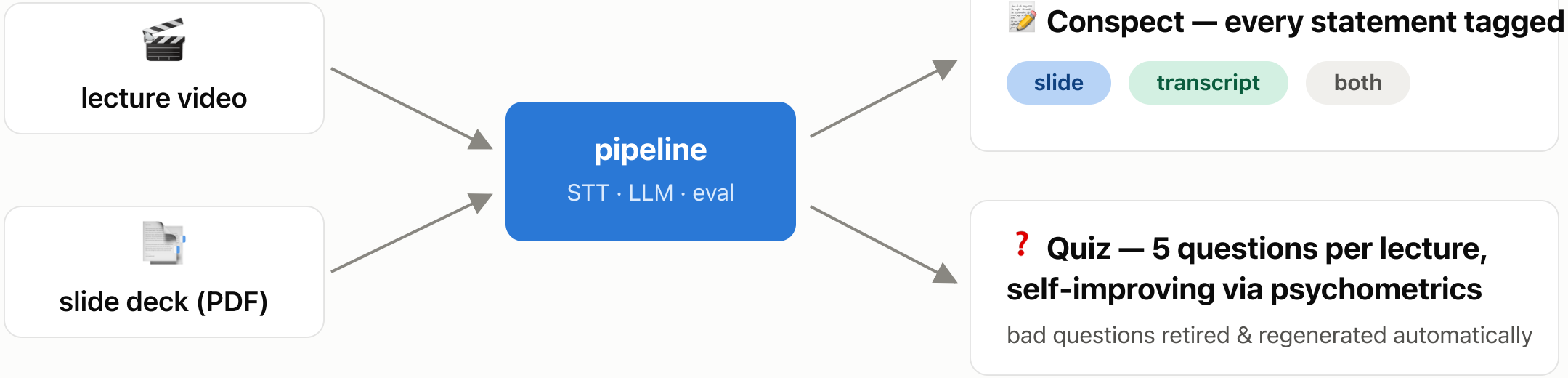
the spoken channel is gone

334 spoken facts of one lecture are not on its slides

Goal: keep what was **said**, not just what was **shown** — and prove it with numbers.

SOLUTION

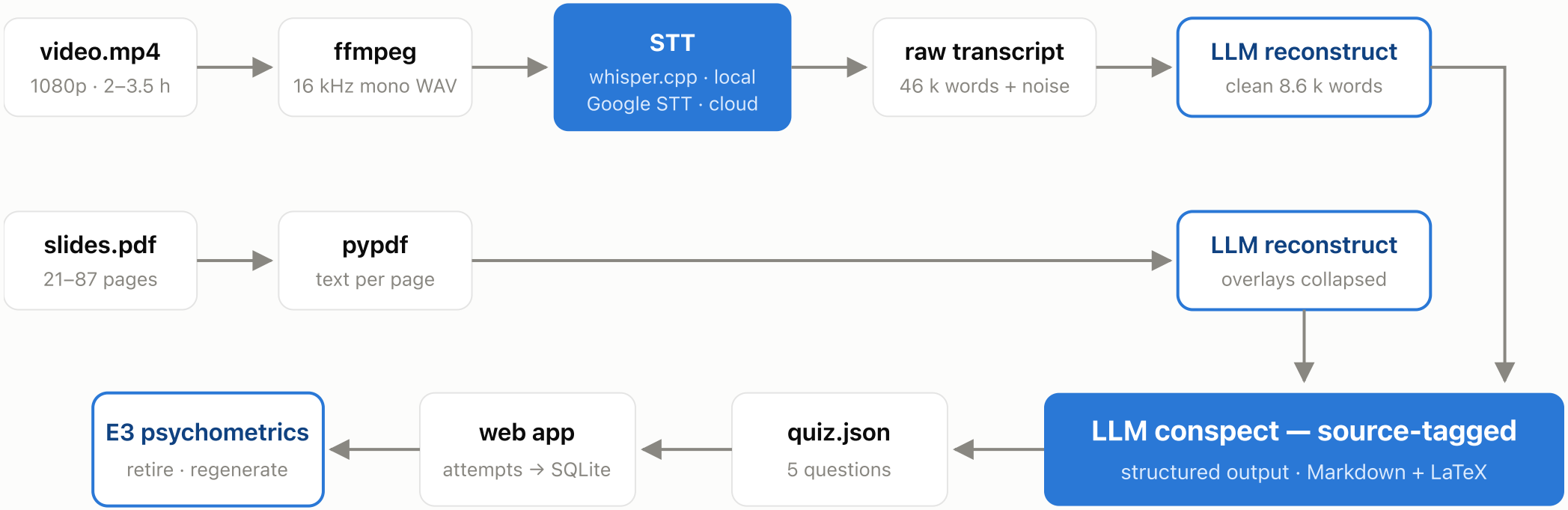
Video + slides → grounded conspect + adaptive quiz



Thesis metric — **transcript-only coverage**: spoken facts absent from slides that survive into the summary. Slides alone score **0%** by definition.

ARCHITECTURE

The pipeline



Two swappable seams — **STT** and **LLM** — everything downstream is provider-independent.

DATA

A real course, collected end to end

"Deep Learning and Computer Vision" — Harbour Space 2026, S. Nikolenko

9
lectures — video + matching slide deck

21.3 h
of 1080p lecture video

619
slide pages (LaTeX/beamer PDF)

19.5 min
golden transcript, hand-made — for WER

40
hand-labelled entailment pairs — judge calibration

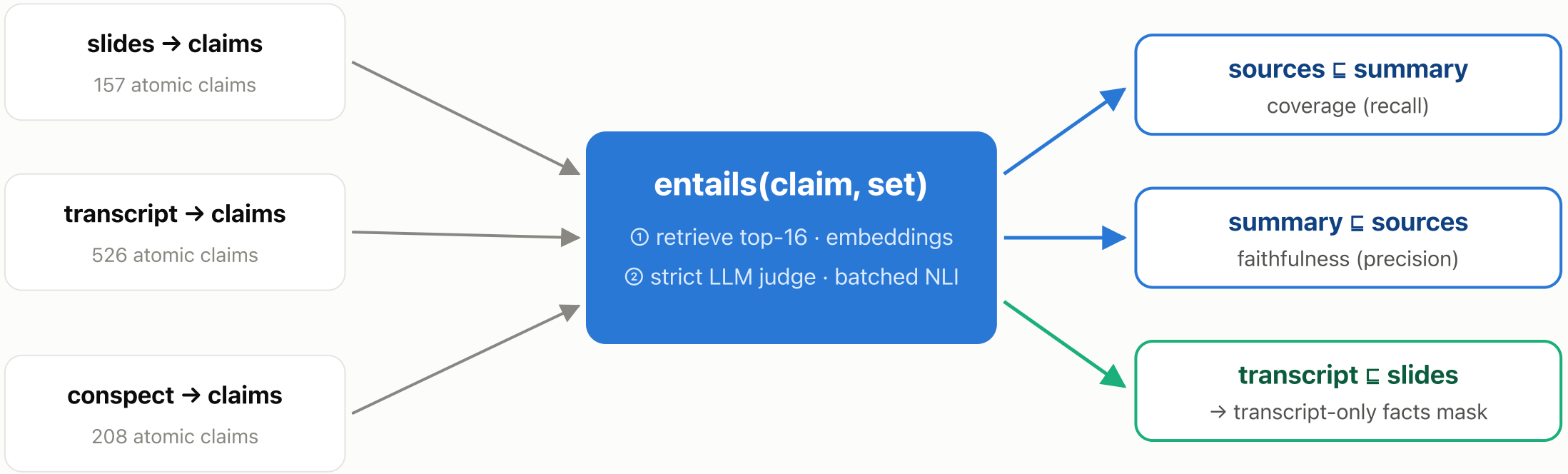
~15
students answering quizzes (E3, in progress)

Dense math content — formulas, theorems, terminology — stresses both STT and summarization.

E1 · METHOD

Evaluate at the level of atomic facts, not words

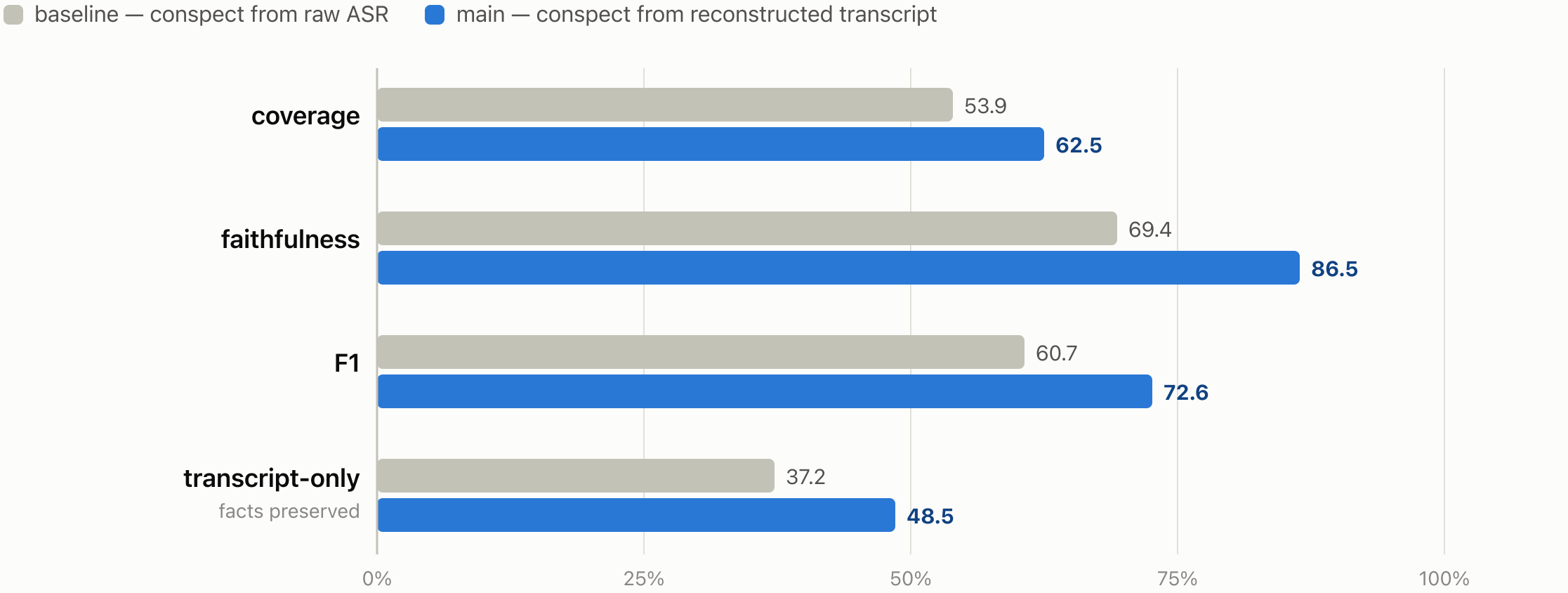
composition · salience: core / supporting / aside



One engine, three directions: **coverage**, **faithfulness** and the **thesis metric** come from the same function.

E1 · RESULTS — LECTURE 6

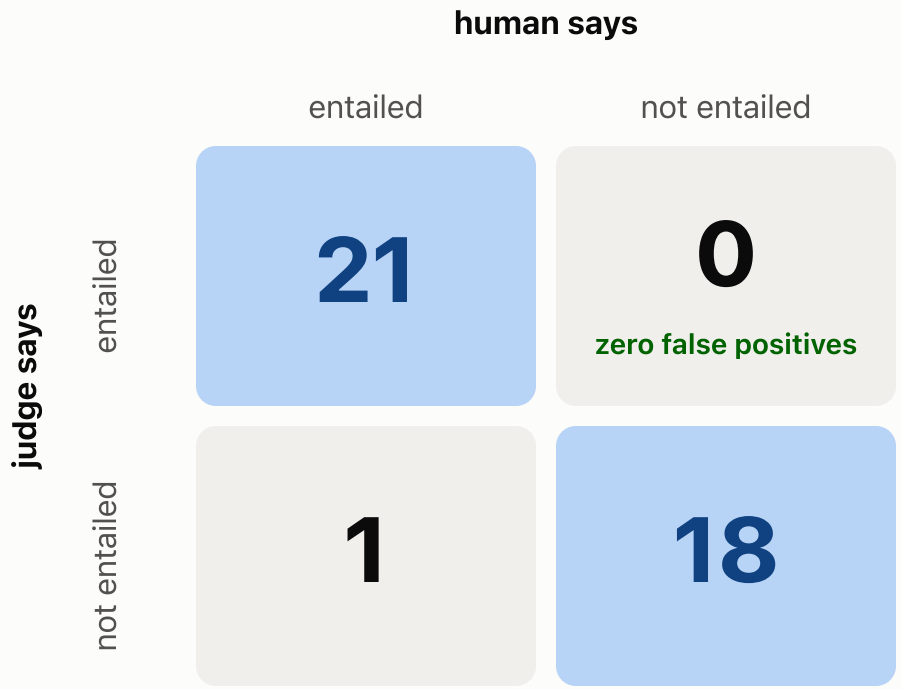
Cleaning the transcript moved every metric



Raw ASR (46 k words) → LLM-reconstructed transcript (8.6 k words, **more** extractable facts: 526 vs 449). F1 **60.7 → 72.6**.

E1 · CAN WE TRUST THE JUDGE?

LLM judge vs 40 hand-labelled pairs



κ = 0.95

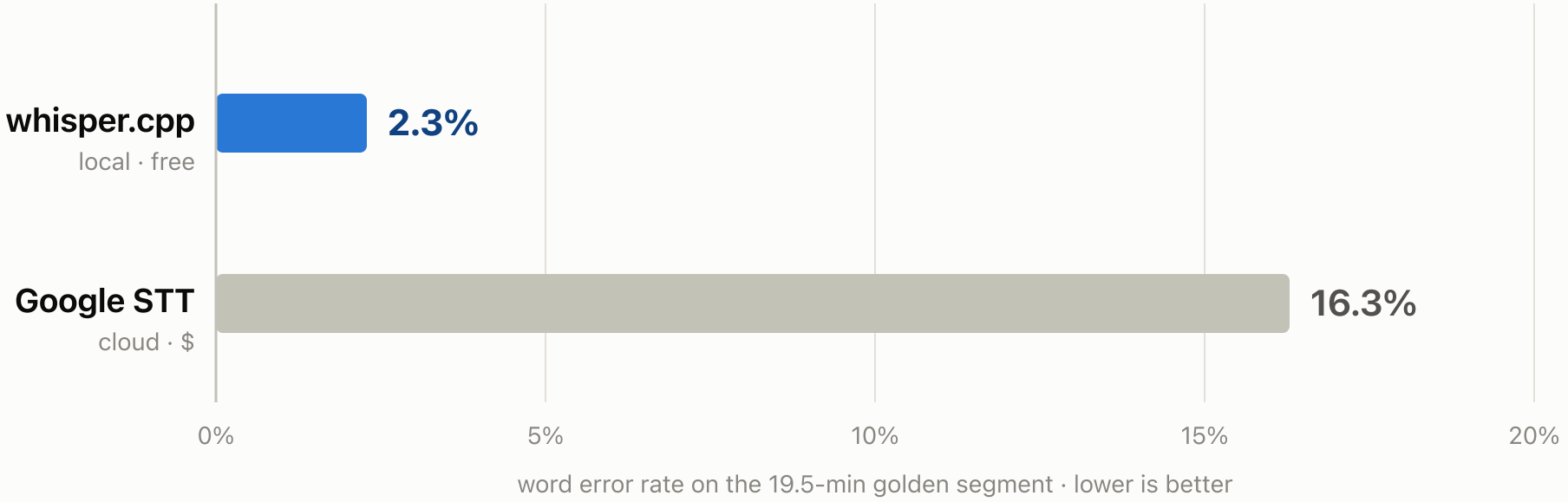
39 / 40 agreement (97.5%)

0 false positives → faithfulness is *not* judge-inflated.
1 miss → coverage numbers are slightly conservative.

The judge is calibrated **before** its numbers are trusted — and recalibrated after any prompt or model change.

E2 · STT BENCHMARK

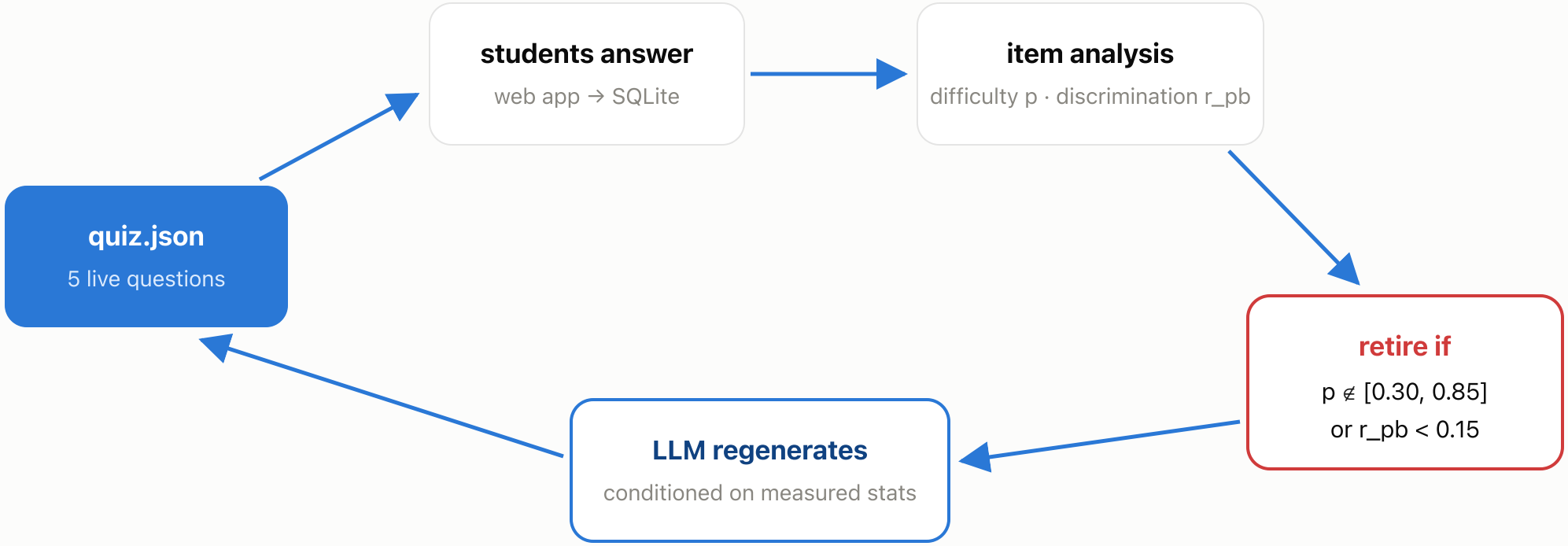
Local whisper.cpp beats cloud STT — 7× lower WER



Pipeline default: **whisper.cpp** — better on accented, terminology-heavy speech, private and $0.

E3 · QUESTION PSYCHOMETRICS

The quiz improves itself



Classical test theory over real attempts — bad questions are **retired, archived and replaced** automatically (n ≈ 15 caveat stated).

STATUS

Done · Next

DONE

- ✓ End-to-end pipeline: STT → reconstruct → conspect → quiz → web
- ✓ E1 engine + results on the dev lecture (F1 72.6%)
- ✓ Judge calibrated: κ = 0.95, 0 false positives
- ✓ E2 WER: whisper.cpp 2.3% vs Google 16.3%
- ✓ E3 item analysis + retire/regenerate loop

NEXT

- → Frozen pipeline over the 8 held-out lectures
- → E2 latency & cost table; TER from slide terms
- → Collect student attempts → run the E3 loop live
- → Ablation: conspect quality vs measurement fairness vs top-k

`github.com/homomorfism/autosummary` · Shamil Arslanov — thank you!